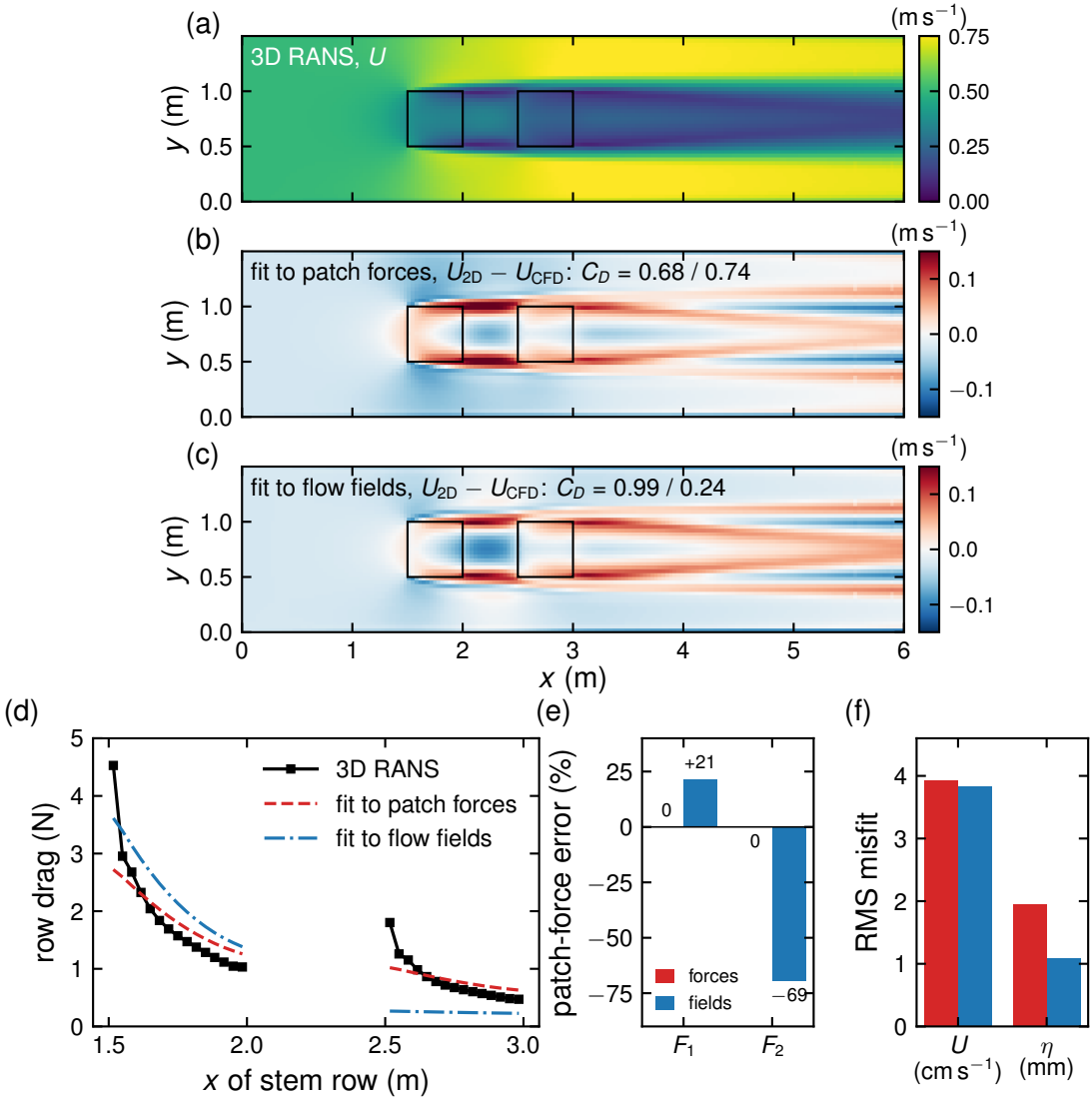

(a)
3D RANS, U
(m s^{-1})
0.75
0.50
0.25
0.00
(b)
fit to patch forces, $U_{2D} - U_{CFD}$: C_D = 0.68 / 0.74
(m s^{-1})
0.1
0.0
−0.1
(c)
fit to flow fields, $U_{2D} - U_{CFD}$: C_D = 0.99 / 0.24
(m s^{-1})
0.1
0.0
−0.1
y (m)
0.0
0.5
1.0
0
1
2
3
4
5
6
x (m)
(d)
row drag (N)
3D RANS
fit to patch forces
fit to flow fields
1.5
2.0
2.5
3.0
x of stem row (m)
(e)
patch-force error (%)
+21
0
0
−69
forces
fields
F_1
F_2
(f)
RMS misfit
U (cm s^{-1})
η (mm)